BUSINESS REQUIREMENTS DOCUMENT (BRD) & COMPREHENSIVE BUSINESS USE CASES

PromptShield: Explainable Generative AI for Scam Detection & Citizen Digital Safety

Project & Governance Parameter	Specification Details
Project Title	PromptShield (Explainable AI Cyber Defense Framework)
Presented By	Team ByteBusters
Lead Developer / Architect	Yashwanth Pathiwada (pathiwadayash@gmail.com)
Document Type	Business Requirements Document (BRD) & Enterprise Business Use Cases
Document Version	v1.0 (Final Evaluator Release Candidate)
Target Evaluation Audience	Academic Evaluation Committee, Technical Jury, Hackathon Evaluators
Repository URL	https://github.com/yashwanthPathiwada/promptshield-ai
Implementation Stack	FastAPI, Python 3.11, React 18, Tailwind CSS, Docker, Uvicorn, GenAI
Release Date	October 2026
Security & Compliance	Zero Server-Side Storage, DPDP Act 2023 Compliant, Local-First Architecture

1. Document Control & Governance

This section maintains formal auditability, stakeholder accountability, and version progression for the PromptShield project in accordance with standard software engineering practices.

1.1 Document Revision History

Version	Release Date	Primary Author	Summary of Revisions	Status
0.1	24-Oct-2026	Team ByteBusters	Initial architectural drafting & heuristic requirements	Superceded
0.5	26-Oct-2026	Yashwanth Pathiwada	Integration of LLM schema specifications & hybrid engine	Superceded
0.9	27-Oct-2026	Team ByteBusters	Addition of 1930 incident complaint workflow & UI specs	Under Review
1.0	29-Oct-2026	Yashwanth Pathiwada	Final Evaluator Edition: Full BRD & 6 Detailed Use Cases	Approved / Ready

1.2 Stakeholder Review & Sign-Off Matrix

Stakeholder Role	Name / Team	Review Responsibility	Sign-Off Status
Project Lead & Architecture	Yashwanth Pathiwada	Core algorithmic logic, FastAPI pipeline & deployment	APPROVED (29-Oct-2026)
Engineering Team	Team ByteBusters	Frontend UI/UX, heuristic rulesets, component testing	APPROVED (29-Oct-2026)
Technical Evaluation Jury	Academic / Hackathon Jury	System completeness, novelty, explainability, compliance	PENDING EVALUATION
Cybersecurity Advisor	Industry Panel	Threat classification alignment & 1930 reporting schema	VERIFIED ACCURATE

> **DOCUMENT PURPOSE FOR EVALUATION**
> This document defines the functional, non-functional, and operational requirements for PromptShield. It serves as the definitive reference for academic evaluators, software engineers, and domain experts to assess the

> completeness, architectural rigor, and real-world societal impact of the solution.

2. Executive Summary & Market Problem

2.1 Executive Overview

PromptShield is a specialized, production-ready, full-stack cybersecurity application engineered to protect digital citizens, students, corporate personnel, and senior citizens from escalating social engineering threats. Unlike conventional spam filters that assign an opaque, uninformative binary label ('Spam' vs. 'Not Spam'), PromptShield bridges the critical 'Explainability Gap' by breaking down exactly why, where, and how a communication attempts to deceive the recipient. It combines an ultra-fast deterministic heuristic engine with cognitive Generative AI to provide color-coded threat scoring, in-context red-flag phrase highlighting, actionable mitigation checklists, and 1-click legal cybercrime reporting.

2.2 The Threat Landscape & The Explainability Gap

- **Explosive Fraud Surge:** According to the Indian Cyber Crime Coordination Centre (I4C), financial cyber fraud exceeded ₹10,000 Crores in recent fiscal cycles, with over 70% initiated via spoofed SMS (smishing), WhatsApp deception, fake job notifications, and urgency-driven bank KYC threats.
- **The Explainability Failure:** Traditional email/SMS spam filters act as opaque black boxes. When a message is flagged, users are rarely told why. Consequently, when scammers modify phrasing to bypass basic keyword filters, uninformed citizens fall prey to psychological triggers such as artificial panic, fear of arrest, or greed.
- **Social Engineering Predominance:** The modern scam vector exploits cognitive human biases (Fear, Urgency, Authority Spoofing, and Scarcity) rather than technical software bugs. Therefore, technical tools must educate the user at the exact moment of exposure to build long-term digital immunity.

2.3 Value Proposition

- **Instant Cognitive Triage:** Evaluates threats via a dual-engine architecture: an instant, zero-cost, regex heuristic engine working in tandem with advanced cognitive LLMs (Gemini, Claude, or GPT).
- **Deep Transparency:** Renders the original message with suspect clauses visually highlighted in context, pairing each flag with an plain-English explanation of the manipulator's intent.
- **Actionable Defense:** Empowers citizens with immediate counter-measures ('Do's & Don'ts') and formats a pre-filled legal complaint ready for submission to the National Cybercrime Portal (cybercrime.gov.in / Helpline 1930).
- **Ethical Privacy Stance:** Operates under a zero-log server architecture. Message contents are analyzed ephemerally in RAM and stored strictly in the client's local browser memory, fully respecting data privacy rights.

3. Vision, Business Objectives & Success KPIs

3.1 Project Vision Statement

To democratize state-of-the-art cognitive cybersecurity intelligence for every digital citizen, transforming vulnerable targets into educated, confident, and proactive first lines of digital defense through transparent, explainable Artificial Intelligence.

3.2 SMART Business Objectives

- **Objective 1 (Real-Time Responsiveness):** Deliver immediate threat classification within 300ms for deterministic heuristics and under 3.5 seconds for deep generative AI reasoning.
- **Objective 2 (Detection Precision):** Achieve greater than 95% detection accuracy across top threat archetypes including Banking KYC, Courier/India Post smishing, Telegram Task jobs, and Digital Arrests.
- **Objective 3 (Explainability Guarantee):** Ensure 100% of detected threats are paired with context-specific, plain-language explanations and practical mitigation steps.
- **Objective 4 (Legal Actionability):** Streamline the incident reporting process from an average of 15 minutes of manual writing down to a single click (under 5 seconds) for cybercrime reporting.
- **Objective 5 (Privacy Compliance):** Guarantee 100% compliance with data privacy mandates by enforcing zero persistent server-side storage of user submissions.

3.3 Key Performance Indicators (KPIs) Target Matrix

Metric / KPI	Current Industry Baseline	PromptShield Target	Evaluation Verification Method
Detection Latency (Heuristic)	1.5s - 3.0s	< 300 ms	FastAPI benchmark timer / automated test suite
Detection Latency (GenAI)	5.0s - 10.0s	< 3.5 s	End-to-end API profiling across test samples
Scam Classification Accuracy	82% - 87% (binary filters)	>= 96.5%	Evaluation across 50 benchmark ground-truth scenarios
Explainability Coverage	< 15% (opaque labels)	100%	Every flagged entity generates an explanatory red-flag card
Incident Report Prep Time	10 to 18 minutes	< 5 seconds	1-Click automated clipboard copy and portal alignment
Server PII Retention	Often retained in cloud logs	0.0% (Zero storage)	Codebase audit: RAM-only processing + local-storage history

4. Stakeholder Analysis & User Personas

4.1 Stakeholder Classification Matrix

Understanding the diverse stakeholder ecosystem ensures that PromptShield addresses operational, academic, and citizen requirements effectively.

Stakeholder Group	Key Interests & Expectations	Primary Impact	RACI Role
Digital Citizens / General Public	Immediate scam verification, simple language, no technical jargon	Prevents direct financial fraud & credential theft	Accountable / User
Senior Citizens & Students	Intuitive one-click UI, protection against panic & job traps	Reduces psychological vulnerability and distress	Informed / User
Academic & Hackathon Jury	Architectural rigor, explainability, schema resilience, novelty	Evaluates project innovation, engineering quality & marks	Consulted / Evaluator
Law Enforcement (1930 / I4C)	Structured, accurate evidentiary complaints with neutralized URLs	Accelerates investigation and freezing of mule accounts	Beneficiary
Engineering Team (ByteBusters)	Clean code, Docker portability, zero-crash fault tolerance	Rapid deployment, maintenance, and future feature scale	Responsible

4.2 Comprehensive User Personas

- **Persona 1 — Rajesh Kumar (Vulnerable Senior Citizen):** A 62-year-old retired bank pensioner. Receives an SMS stating: 'Your SBI Yono account will be blocked in 24 hours. Update PAN card immediately at http://bit.ly/sbi-pan-kyc'. He experiences intense panic, fearing loss of pension access. With PromptShield, he pastes the SMS, sees an instant RED Risk Score (94/100), reads a clear red-flag explaining 'Banks never send bit.ly shortlinks or demand urgent PAN uploads via SMS', and follows the checklist to block the number.
- **Persona 2 — Ananya Sharma (Aspiring Student):** A 20-year-old college undergraduate looking for pocket money. Receives a WhatsApp offer: 'Work from home! Earn ₹3,000/day simply by liking YouTube videos and subscribing to Telegram channels'. Unsure if legitimate, she runs PromptShield. The app highlights the advance-fee trap, warns that 'Paying a deposit to unlock earned commissions is a textbook task scam', and prevents her from transferring her semester savings.
- **Persona 3 — Vikram Malhotra (Busy Professional):** A 34-year-old marketing executive managing dozens of parcel deliveries weekly. Receives an SMS: 'India Post: Package delivery failed due to incomplete address. Update within 12h or return to sender: indiapost-redirection.com'. He uses PromptShield to verify if it is legitimate before clicking. PromptShield flags the spoofed domain while contrasting it with India Post's official `.gov.in` domain.
- **Persona 4 — Inspector S. Rao (Cybercrime Cell Officer):** An investigating sub-inspector receiving citizen grievances. Complains that victims usually provide vague verbal statements without the exact message text,

URL, or sender header. With PromptShield's 1-Click Cybercrime Report, the officer receives standardized, time-stamped evidentiary complaints, significantly cutting preliminary inquiry time.

5. Scope Management (Boundary Definition)

5.1 In-Scope Capabilities

- **Multi-Vector Text Analysis:** Support for pasting raw SMS, email content, job descriptions, chat logs, and delivery notifications.
- **Dual-Engine Pipeline:** Deterministic heuristic analyzer containing rules for urgency, financial triggers, credential traps, shortlinks, and telecom sender prefixes.
- **Multi-LLM Semantic Ingestion:** Generative AI cognitive analyzer supporting Gemini 1.5 Flash, Claude 3.5 Sonnet, and OpenAI GPT-4o-mini with Pydantic JSON schema enforcement.
- **Dynamic Risk Quantification:** Visual 0 to 100 risk scale divided into Safe (0–30), Suspicious (31–69), and Likely Scam (70–100).
- **Interactive Contextual Highlighter:** Visual highlight of deceptive phrases paired with plain-language threat explanations.
- **Actionable Safety Protocol:** Interactive checklist specifying mandatory immediate actions ('What To Do') and critical prohibitions ('What NOT To Do').
- **1-Click 1930 Incident Formatter:** Pre-fills formatted evidentiary complaints aligned with India's National Cybercrime Portal standards.
- **Local-First Privacy History:** Stores scan history strictly in browser localStorage without server persistence.

5.2 Out-of-Scope Capabilities (Explicit Boundaries)

- **Direct Banking Transactions:** PromptShield is an analytical and educational defense tool; it does not directly freeze bank accounts or reverse UPI transactions.
- **Offensive Cyber Warfare:** PromptShield does not conduct counter-attacks, port scanning, or offensive intrusion against scam servers.
- **Passive Carrier Eavesdropping:** PromptShield does not tap or passively eavesdrop on cellular voice or carrier networks.

5.3 Phased Delivery Roadmap

Phase	Milestone Target	Key Deliverables	Status
Phase 1 (MVP)	Web Application & Core Dual Engine	FastAPI backend, React 18 UI, 0-100 meter, red flags, 1930 formatter	COMPLETE / EVALUATOR READY
Phase 2 (Near-Term)	Browser Extension & Mail Integration	One-click analysis inside Gmail, Outlook webmail, and WhatsApp Web	Q1 2027 Roadmap

Phase 3 (Enterprise)	Android On-Device SMS Scanner & Voice	Background SMS reader with Hindi, Telugu, Tamil, and Bengali audio explanations	Q2 2027 Roadmap

6. Functional Requirements (FR) with MoSCoW Prioritization

The functional requirements specify the complete behavioral capabilities of the system. Each requirement is mapped to its MoSCoW priority, user story, and formal acceptance criteria.

Req ID	Requirement Name	Priority	Functional Specification	Acceptance Criteria
FR-01	Text Ingestion & Sanitization	Must Have	Accepts text up to 10,000 characters from web UI. Strips hidden zero-width control chars.	Rejects empty or <3 char inputs with HTTP 400. Handles emojis & multi-line inputs safely.
FR-02	Deterministic Rule Engine	Must Have	Regex & heuristic analyzer scanning for 40+ fraud indicators (KYC, urgent deadlines, APKs, shortlinks).	Executes in < 50ms without external network calls; produces baseline risk score.
FR-03	Cognitive LLM Semantic Engine	Must Have	Integrates Gemini/Claude/OpenAI with system prompts enforcing JSON structured output.	Produces structured JSON containing risk_score, category, summary, red_flags, and checklist.
FR-04	Schema Validation & Fallback Guardrails	Must Have	Validates LLM outputs against Pydantic schema. Falls back to heuristic engine on parse failure.	Zero crashes if an LLM outputs malformed text or API times out.
FR-05	Dynamic 0-100 Risk Scoring	Must Have	Normalizes threat score: Safe (0-30, Green), Suspicious (31-69, Amber), Likely Scam (70-100, Red).	Color gauge and badge dynamically update in UI according to score threshold.
FR-06	In-Context Red Flag Highlighter	Must Have	Matches returned deceptive quotes against original text and renders colored interactive spans.	Clicking or hovering over flagged span renders the exact manipulation explanation.
FR-07	Actionable Safety Protocol	Must Have	Generates dynamic 'What To Do' and 'What NOT To Do' checklists tailored to threat	Checkboxes are interactively toggleable by the citizen in real-time.

			category.	
FR-08	1-Click Incident Complaint Formatter	Should Have	Assembles incident summary, sender, flagged URLs, and timestamps into standardized 1930 report.	One-click button copies complete complaint text to system clipboard.
FR-09	Privacy-Preserving History Drawer	Should Have	Persists scan results in browser localStorage. Allows filtering, viewing, and clearing.	No user message or scan history is ever stored in backend databases or server files.
FR-10	Scam Awareness Knowledge Base	Should Have	Interactive educational modules covering top scam patterns (KYC, Job, UPI, Courier, Digital Arrest).	Expandable educational cards render with examples, red flags, and preventive tips.
FR-11	One-Click Quick Presets	Could Have	Pre-loads real-world scam and genuine test cases for instant evaluation and demo testing.	Clicking a sample populates input and triggers analysis immediately.

7. Non-Functional Requirements (NFR) & Quality Attributes

NFR ID	Quality Dimension	Target Quantitative Metric / SLA	Engineering Architecture & Implementation Strategy
NFR-01	Performance & Latency	Deterministic Heuristic: < 250ms LLM Semantic Inference: < 3.5s	Pre-compiled regex patterns, async FastAPI pipeline, lightweight non-blocking event loop.
NFR-02	Availability & Uptime	99.9% application uptime with graceful degradation	Dual-engine architecture: If cloud LLM API is unreachable, deterministic engine serves instant verdict.
NFR-03	Information Security	OWASP Top 10 compliance, zero script injection	Content sanitization on ingest, neutralizing active hyperlinks, strict CORS headers.
NFR-04	Data Privacy & Compliance	100% adherence to DPDP Act 2023 & GDPR principles	Ephemeral in-memory processing only. Zero persistent databases, cookies, or tracking pixels.

NFR-05	Usability & Accessibility	WCAG 2.1 Level AA compliance, responsive layout	Tailwind CSS responsive design, mobile-first layouts, color-blind friendly badges, high contrast text.
NFR-06	Scalability & Concurrency	Support 500+ concurrent requests without degradation	Stateless ASGI Uvicorn workers wrapped in multi-stage Docker container for horizontal auto-scaling.
NFR-07	Portability & DevOps	Single-command deployment anywhere	Multi-stage Dockerfile bundling Vite React frontend and Python FastAPI backend into a single image.

8. Comprehensive Business Use Cases (BUC)

This section contains 6 fully realized, industry-standard Business Use Cases detailing operational workflows, edge cases, and business value for academic evaluators.

BUC-01: Analysis of Spoofed Bank KYC / PAN Suspension Smishing SMS

Use Case Parameter	Specification
Use Case ID & Title	BUC-01: Analysis of Spoofed Bank KYC / PAN Suspension Smishing SMS
Primary Actor	Everyday Citizen / Senior Citizen (e.g., Rajesh Kumar)
Trigger Event	User receives an alarming SMS: 'Dear Customer, your SBI account will be blocked today. Update PAN immediately at http://bit.ly/sbi-pan-update'.
Pre-Conditions	User has access to PromptShield via desktop or mobile browser.
Main Success Scenario (Flow)	1. User navigates to PromptShield and pastes the suspicious SMS text into the input field. 2. User clicks 'Analyze Message' (or presses Enter). 3. System checks text validity and initiates parallel dual-engine analysis. 4. Deterministic engine flags urgent deadline ('blocked today'), shortlink ('bit.ly'), and banking trigger ('PAN'). 5. Cognitive LLM evaluates semantic intent: identifies credential harvesting impersonating SBI. 6. System computes final Risk Score of 94/100 (Red - Likely Scam). 7. UI renders highlighted text: 'blocked today' (Urgency Pressure), 'http://bit.ly/...' (Credential Phishing Link). 8. UI displays customized checklist: Do NOT click link; Do NOT enter OTP; Verify directly in official YONO app.

	9. Result is cached into user's browser Local History.
Alternative / Exception Flows	4a. Cloud LLM rate limit or timeout: System automatically falls back to Deterministic Heuristic Core, scoring threat at 88/100 with offline rule explanations. 5a. Input message is less than 3 characters: System prompts user to enter complete message.
Post-Conditions	User understands the exact scam mechanism, does not click the malicious link, and avoids financial loss.
Business Value & Evaluator Takeaway	High financial loss prevention; educates the citizen on banking safety norms and eliminates panic-driven compliance.

BUC-02: Detection of Telegram / YouTube 'Part-Time Task' Job Scam

Use Case Parameter	Specification
Use Case ID & Title	BUC-02: Detection of Telegram / YouTube 'Part-Time Task' Job Scam
Primary Actor	University Student / Job Seeker (e.g., Ananya Sharma)
Trigger Event	User receives an unsolicited WhatsApp message offering ₹5,000/day for 'liking videos and subscribing to Telegram channels'.
Pre-Conditions	User is evaluating the job offer and seeks verification before sending money or personal details.
Main Success Scenario (Flow)	1. User pastes WhatsApp job offer into PromptShield. 2. System analyzes message linguistics and recruitment tactics. 3. Heuristic engine flags phrases: 'like YouTube videos', 'daily salary ₹5000', 'join Telegram group'. 4. LLM analyzes social engineering: identifies advance-fee / task pyramid fraud pattern. 5. System assigns Risk Score of 88/100 (Red - Likely Scam). 6. Explainer warns: 'Scammers pay minor amounts (₹150) initially to build trust, then demand deposit to unlock earnings'. 7. Action checklist instructs: Block sender; Never pay money to receive job wages; Report WhatsApp number.
Alternative / Exception Flows	3a. Message contains obfuscated characters (e.g., 'T-e-l-e-g-r-a-m'): Sanitization engine normalizes text before matching rules.

Post-Conditions	User realizes the deceptive nature of the offer, declines to join the Telegram group, and avoids losing money.
Business Value & Evaluator Takeaway	Protects young job seekers from multi-thousand-rupee task fraud; builds awareness of modern social engineering.

BUC-03: Verification of Legitimate Courier Delivery Notification (False Positive Prevention)

Use Case Parameter	Specification
Use Case ID & Title	BUC-03: Verification of Legitimate Courier Delivery Notification (False Positive Prevention)
Primary Actor	E-Commerce Shopper / Corporate Professional (e.g., Vikram Malhotra)
Trigger Event	User receives an automated SMS: 'Your BlueDart parcel #7392819 will be delivered today by courier agent Ramesh (Ph: 9876543210). Track at bluedart.com'.
Pre-Conditions	User wants to confirm the message is genuine before contacting the delivery agent.
Main Success Scenario (Flow)	1. User pastes courier message into PromptShield. 2. Deterministic engine verifies official domain (`bluedart.com`), absence of threats or urgent payment demands. 3. Cognitive LLM evaluates context: Informational delivery alert without credential demands or coercive language. 4. System computes Risk Score of 10/100 (Green - Safe). 5. UI renders Safe badge with summary: 'Legitimate transactional communication. No deceptive triggers detected'. 6. Safety checklist advises: Verify tracking number directly on official carrier portal.
Alternative / Exception Flows	2a. URL contains deceptive lookalike domain (e.g., 'bluedart-delivery-fee.com'): System switches verdict to Likely Scam (Score 91).
Post-Conditions	User confidently receives delivery without confusion or undue suspicion, validating system reliability.
Business Value & Evaluator Takeaway	Critical benchmark for evaluators: Proves PromptShield does not aggressively false-flag legitimate business communications.

BUC-04: Detection of Electricity Bill Disconnection Extortion Threat

Use Case Parameter	Specification
Use Case ID & Title	BUC-04: Detection of Electricity Bill Disconnection Extortion Threat
Primary Actor	Small Business Owner / Resident
Trigger Event	User receives an SMS at 7:00 PM: 'Electricity power will be disconnected at 9:30 PM tonight due to unpaid bill. Immediately contact Electricity Officer at 9123456789'.
Pre-Conditions	User is startled by the imminent power cut threat and considers calling the private number.
Main Success Scenario (Flow)	1. User pastes the disconnection threat into PromptShield. 2. Heuristic engine flags extreme urgency ('tonight 9:30 PM'), threat of disconnection, and unofficial personal phone number. 3. LLM identifies standard Indian utility scam vector (impersonating state electricity distribution boards). 4. System assigns Risk Score of 92/100 (Red - Likely Scam). 5. UI highlights: 'disconnected at 9:30 PM' (Coercive Urgency) and '9123456789' (Unofficial Private Mobile Contact). 6. Action checklist clarifies: Government power boards never disconnect power via personal mobile SMS or ask for remote access apps (AnyDesk/TeamViewer).
Alternative / Exception Flows	1a. User clicks 'Apple Quick Preset' for Electricity scam: System auto-populates scenario for instant evaluation.
Post-Conditions	User avoids calling the scammer, checks bill on official electricity utility app, and discovers zero arrears.
Business Value & Evaluator Takeaway	Neutralizes widespread extortion tactics targeting residential and commercial utility consumers.

BUC-05: One-Click Generation of Official Cybercrime Complaint for Helpline 1930

Use Case Parameter	Specification
Use Case ID & Title	BUC-05: One-Click Generation of Official Cybercrime Complaint for Helpline 1930

Primary Actor	Scam Target / Active Citizen Reporting Fraud
Trigger Event	User has analyzed a confirmed scam message and wants to file a complaint with national cyber authorities.
Pre-Conditions	PromptShield has completed analysis and classified message as Suspicious or Likely Scam.
Main Success Scenario (Flow)	1. User reviews analysis results and scrolls to 'Official Cybercrime Report' section. 2. System automatically populates a structured complaint template containing: - Incident Classification (e.g., Banking KYC Phishing / Task Fraud) - Sender Identification & Timestamp - Neutralized Evidence Text - Extracted Malicious URLs & Phone Numbers - PromptShield Risk Assessment (Score: 94/100) 3. User clicks 'Copy Official Incident Complaint'. 4. System copies formatted text to clipboard and displays direct links to National Cybercrime Portal (`cybercrime.gov.in`) and National Helpline (`1930`). 5. User pastes formatted text into portal grievance form in seconds.
Alternative / Exception Flows	3a. Browser clipboard permissions blocked: System displays complaint in selectable read-only text area with instructions.
Post-Conditions	Official authorities receive high-quality, standardized digital evidence, expediting investigation.
Business Value & Evaluator Takeaway	Dramatically lowers friction of reporting cybercrime from 15 minutes of manual writing to 5 seconds.

BUC-06: Autonomous Heuristic Fallback During Cloud LLM / Network Outage

Use Case Parameter	Specification
Use Case ID & Title	BUC-06: Autonomous Heuristic Fallback During Cloud LLM / Network Outage
Primary Actor	Evaluation Jury / Offline User / Developer
Trigger Event	Cloud LLM provider (Gemini/OpenAI/Claude) experiences an API outage or local network connectivity is severed.
Pre-Conditions	User submits a message during an active upstream API failure or rate limit condition.

Main Success Scenario (Flow)	1. User submits suspicious text for analysis. 2. Backend FastAPI service attempts to invoke configured LLM provider. 3. API call raises timeout (HTTP 504 / ConnectionError / 429 Rate Limit). 4. Built-in Circuit Breaker activates without crashing the server. 5. Request is routed immediately to the Deterministic Offline Heuristic Engine. 6. Heuristic engine evaluates regex rules, linguistic tokens, and URL threat lists. 7. System constructs complete `AnalyzeResponse` schema with fallback flag `is_fallback: true`. 8. UI displays accurate risk score, highlighted flags, and safety checklist seamlessly with zero downtime.
Alternative / Exception Flows	5a. Backend itself is offline: Frontend includes offline analyzer JS fallback in browser.
Post-Conditions	The application remains 100% functional and responsive even with zero internet or API keys.
Business Value & Evaluator Takeaway	Guarantees rock-solid reliability for live hackathon demos and evaluator grading.

9. System Architecture & Technical Specifications

9.1 Dual-Engine Architecture Pipeline

PromptShield is architected around a dual-engine processing pipeline combining low-latency rule engines with deep semantic LLMs:

- **1. Ingestion & Sanitization Layer:** Pre-processes raw text, normalizes Unicode lookalikes, strips zero-width spaces, and extracts candidate URLs.
- **2. Deterministic Rule Core:** Executes 40+ pre-compiled regular expressions identifying urgency triggers, banking terms, shortlinks, telecom headers, and APK file patterns in < 50ms.
- **3. Cognitive Semantic LLM:** Dispatches context to Gemini 1.5 Flash / Claude 3.5 Sonnet / GPT-4o-mini using strict system prompts enforcing structured JSON schema.
- **4. Pydantic Schema Validator:** Validates response against Pydantic models. If schema validation fails or an exception occurs, fallback rules take over instantaneously.
- **5. Explainability & Output Formatter:** Maps extracted threat quotes to original text indices, generating interactive red-flag spans, checklists, and 1930 complaint drafts.

9.2 Technical Stack Specification

Layer / Component	Technology / Framework	Version	Architectural Role & Justification

Frontend UI	React.js + Tailwind CSS	React 18.3 / Tailwind 3.4	Declarative, responsive citizen UI with accessible state management
Build Tool	Vite	Vite 5.x	Sub-second hot-reloading and optimized production bundle
Icons	Lucide React	Lucide 0.380+	Crisp, intuitive visual indicators for threats and safety actions
Backend API	FastAPI (Python)	FastAPI 0.111+ / Python 3.11+	High-performance asynchronous REST endpoints with native Pydantic validation
ASGI Server	Uvicorn	Uvicorn 0.30+	Lightning-fast ASGI event loop capable of handling thousands of requests
AI Orchestration	Multi-Provider SDKs	Google GenAI / Anthropic / OpenAI	Cognitive semantic analysis with zero-shot explainable red flags
Containerization	Docker (Multi-Stage)	Docker Engine 26+	Multi-stage build packaging React dist and Python FastAPI in one container
Deployment	Render / Railway Cloud	Cloud PaaS	Automatic CI/CD pipeline triggered upon GitHub push

10. Risk Assessment & Mitigation Matrix

Identifying technical and operational risks ensures that PromptShield maintains resilience in production environments.

Risk ID	Risk Description	Severity	Likelihood	Risk Level	Proactive Mitigation Strategy
R-01	LLM Hallucination or Unstructured Format	High	Low	Medium	Enforced strict Pydantic JSON schema; automatic heuristic fallback if output fails validation.
R-02	Third-Party Cloud AI API Outage / Rate Limit	High	Medium	High	Built-in offline heuristic engine activates seamlessly;

					app functions 100% offline.
R-03	Zero-Day Obfuscated Scam Phrasing	Medium	Medium	Medium	Dual-engine captures intent: even if keywords change, semantic LLM identifies social engineering.
R-04	False Positive on Urgent Legitimate Alert	Medium	Low	Low	Balanced scoring algorithm requiring multiple corroborating threat vectors before flagging RED.
R-05	User Privacy Leak of Pasted Sensitive Data	Critical	Low	Low	Zero server-side persistence; no database; all scan history resides solely in browser localStorage.

11. Verification, Acceptance Criteria & Formal Sign-Off

11.1 Evaluator Rubric Alignment

PromptShield is specifically engineered to achieve maximum marks across standard academic and hackathon evaluation criteria:

- **1. Innovation & Novelty (25%):** Transitions cybersecurity from opaque black-box labels to transparent, educational, in-context cognitive explainability.
- **2. Technical Execution & Architecture (25%):** Full-stack integration featuring FastAPI, React 18, Vite, Docker, strict Pydantic schemas, and multi-model support.
- **3. Societal Impact & Relevance (25%):** Directly targets India's ₹10,000+ Cr cyber fraud epidemic; integrates with National Helpline 1930 and cybercrime.gov.in.
- **4. Completeness & Live Demonstration (25%):** Fully working live application with sample test presets, zero-crash heuristic fallback, and responsive UI design.

11.2 Formal Evaluator Sign-Off

Evaluation Role	Evaluator Name / Jury Member	Assessment Verdict	Signature / Date
Lead Project Evaluator	_____	[ ] Outstanding [ ] Approved	_____ / _____ / 2026
Technical Architecture Jury	_____	[ ] Outstanding [ ] Approved	_____ / _____ / 2026

Cybersecurity Domain Expert	_____	[ ] Outstanding [ ] Approved	_____ / _____ / 2026
Academic Committee Chair	_____	[ ] Outstanding [ ] Approved	_____ / _____ / 2026

SUBMITTED BY TEAM BYTEBUSTERS

Document compiled and verified by Team ByteBusters (Lead Developer: Yashwanth Pathiwada).
Official Project Repository: https://github.com/yashwanthPathiwada/promptshield-ai
Thank you to the evaluation committee for reviewing this submission.